

together, the two modules are called CSM; and the lunar module (LM), to take two of the crew to the lunar surface, support them on the Moon, and return them to the CSM in lunar orbit.

When attached together, the total length of the Command and Service Module (CSM) was 11 metres with a maximum diameter of 3.9 metres. Including propellants and expendables, the launch mass was approximately 28,800 kg, of which the Command Module had a mass of over 5550 kg and the Service Module over 23,000 kg.

Telecommunications included voice, television, data, and tracking and ranging subsystems for communications between astronauts, CM, LM, and Earth. Voice contact was provided by an S-band uplink and downlink system. Tracking was done through a unified S-band transponder. A high gain steerable S-band antenna consisting of four 79-cm diameter parabolic dishes was mounted on a folding boom at the aft end of the SM. Two VHF scimitar antennas were also mounted on the SM. There was also a VHF recovery beacon mounted in the CM. The CSM environmental control system regulated cabin atmosphere, pressure, temperature, carbon dioxide, odours, particles, and ventilation and controlled the temperature range of the electronic equipment.

The Command Module was a conical pressure vessel with a maximum diameter of 3.9 m at its base and a height of 3.65 m. It was made of an aluminium honeycomb sandwich bonded between sheet aluminium alloy. The base of the CM consisted of a heat shield made of brazed stainless steel honeycomb filled with a phenol epoxy resin as an ablative material and varied in thickness from 1.8 to 6.9 cm. At the tip of the cone was a hatch and docking assembly designed to mate with the lunar module.

The Command Module was divided into three compartments. The forward compartment in the nose of the cone held the three 25.4-metre diameter main parachutes, two 5-metre drogue parachutes, and pilot mortar chutes for Earth landing. The aft compartment was situated around the base of the CM and contained propellant tanks, reaction control engines, wiring, and plumbing. The crew compartment comprised most of the volume of space. Three astronaut couches were lined up facing forward in the centre of the compartment. A large access hatch was situated above the centre couch. A short access tunnel led to the docking hatch in the nose. The crew compartment held the controls, displays, navigation equipment and other systems used by the astronauts. The CM had five windows: one in the access hatch, one next to each astronaut in the two outer seats, and two forward-